

operating conditions make fixes expensive and degrade overall system performance. Early-stage, ‘left-shifted’ validation is essential to identify issues before they become costly system-level failures.

• **Infrastructure** remains another weak link. Limited access to affordable prototyping, testing, and certification labs restricts grassroots innovation. As a result, promising ideas often remain theoretical and fail to transition into production-ready systems.

Together, these gaps highlight a crucial reality: building a semiconductor ecosystem is not just about funding or fabrication units. It requires deep systemic alignment across skills, tools, processes, and infrastructure. Addressing these foundational challenges will be key to transforming India from a participant in the global electronics landscape into a true leader.

Why silicon alone is not enough

A key realisation emerging from India’s semiconductor journey is both surprising and transformative: silicon itself accounts for only a small fraction, roughly 10%, of a product’s overall success. The remaining 90% is determined by everything that surrounds the chip, how it is integrated, validated, adapted, and deployed in real-world environments. This insight fundamentally shifts the narrative from a narrow focus on chip fabrication to a broader, system-level perspective.

Even if India successfully establishes fabrication units (fabs) and ATMP (assembly, testing, marking, and packaging) facilities, that alone does not guarantee success. The real challenge begins after the silicon is ready.

At this stage, system architecture design becomes crucial. Without a well-defined architecture tailored to specific use cases—whether automotive, industrial, or defence—the chip cannot deliver meaningful value. Architecture determines how

The missing layer: System-level thinking

India often focuses on component-level development, but the real need is end-to-end system ownership. This includes:

- Architecture design
- Hardware-software co-design
- Validation pipelines
- Deployment strategies

different components interact, how efficiently resources are utilised, and how well the system performs under real-world constraints.

Equally important is the development of robust validation ecosystems. A chip that performs well in controlled lab conditions may fail when exposed to practical environments involving temperature variations, electromagnetic interference, mechanical stress, or inconsistent power conditions. Comprehensive validation frameworks covering reliability, safety, and compliance are essential to ensure products are not only functional but also dependable at scale.

Beyond the hardware, application-layer development plays a defining role in shaping user experience and functionality. Software frameworks, middleware, and domain-specific applications must be tightly aligned with the underlying hardware to unlock its full potential. Without this layer, even the most advanced silicon remains underutilised.

Finally, manufacturing readiness is a decisive factor in bridging the gap between prototype and production. This includes ensuring consistent quality, scalable production processes, supply chain coordination, and adherence to industry standards. A design that works in the prototype stage may encounter unforeseen challenges when scaled to mass production, particularly in diverse and demanding conditions such as those found in India.

Taken together, this perspective underscores a critical shift: success

in the semiconductor domain is not defined by the chip alone, but by the strength of the entire ecosystem built around it.

Accelerators: The untapped opportunity

As India aspires to strengthen its semiconductor ecosystem, a strategic question arises: Where should it compete first?

Entering highly complex and capital-intensive domains such as CPUs and GPUs, dominated by global giants, may not be the most pragmatic starting point. Instead, a more focused, high-impact pathway lies in developing specialised hardware accelerators.

Accelerators are purpose-built chips designed to execute specific tasks with high efficiency. Unlike general-purpose processors, they are optimised for targeted applications such as EV motor control, functional safety systems, edge AI inference, signal processing, and hardware-based security modules.

These domains are rapidly expanding, particularly in sectors such as automotive, industrial automation, and smart infrastructure, where India already has a growing market presence.

Despite their importance, India currently lacks a strong foothold even in accelerator design and deployment. This gap, however, presents a unique opportunity. Compared to CPUs and GPUs, accelerators offer a lower barrier to entry, require smaller teams, and can be developed with domain-specific expertise.

More importantly, they allow Indian companies to solve localised problems—such as optimising for Indian road conditions, power environments, or industrial use cases—while gradually building deeper capabilities in silicon design.

In essence, accelerators provide a practical and scalable entry point into the semiconductor value chain, enabling India to create immediate